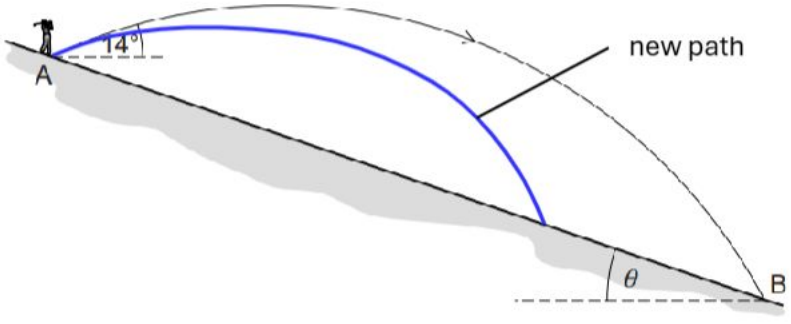
1(a)(i)	horizontal component $= 67\cos 14^\circ$ $= 65.01$ $\approx 65.0 \text{ m s}^{-1}$ vertical component $= 67\, sin\, 1\, 4^\circ$ $= 16.21$ $\approx 16.2 \text{ m s}^{-1}$ (accept 2 or 3 SF)	**A1**
1(a)(ii)	horizontal displacement $= 65.01 \times 4.9$ $= 318.5$ $\approx 319 \text{ m}$ (accept 2 or 3 SF)	**C1** **A1**
1(a)(iii)	Using $s_y = u_y t + \frac{1}{2} a_y t^2$, Taking downward as positive, $s_y = (-16.21)(4.9) + \frac{1}{2}(9.81)(4.9)^2$ $= 38.34$ $\approx 38.3 \text{ m}$ (accept 2 or 3 SF)	**C1** **A1**
1(a)(iv)	angle of the slope to the horizontal $\theta = tan^{-1}\left(\frac{38.34}{318.5}\right)$ $= 6.864$ $\approx 6.86^\circ$ (accept 2 or 3 SF)	**C1** **A1**
1(b)(i)	 [1] peak lower and nearer [1] new path under present path and shape is not symmetrical	**A1** **A1**

1(b)(ii)1.	The path of the ball at the start is determined by the direction of the force of the club acting on the ball **or** air resistance has acted only for a short period of time.	**B1**